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Driving assistance system and method

Abstract

A driving assistance system for a work vehicle may comprise a service brake configured to decelerate the work vehicle upon actuation of the service brake, a parking brake and a transmission including a clutch. A controller may be configured to disengage the clutch when the work vehicle speed is less than a speed threshold and when the service brake is actuated, whereby the work vehicle is brought to a stop.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to European Patent Application No. EP22305270 filed on Mar. 10, 2022.

TECHNICAL FIELD

(2) The present disclosure relates to the field of driving assistance, and more particularly to a driving assistance system for a work vehicle such as a tractor.

BACKGROUND

(3) Work vehicles should meet safety and productivity requirements in all the tasks to which they are assigned. For instance, when a work vehicle is stopped on a slope, it should be able to resume moving in a simple and reliable manner. This includes, among others, reducing or even preventing rollback.

(4) In this perspective, hill start assistance systems have been proposed, such as in U.S. Pat. No. 10,316,959 to Pawel Kuros. However, these systems often require sophistications of the components of the work vehicle, which may make them complex and expensive.

SUMMARY

(5) The object of the present invention is to at least substantially remedy the above-mentioned drawbacks.

(6) A driving assistance system for a work vehicle may comprise a service brake configured to decelerate the work vehicle upon actuation of the service brake, a parking brake and a transmission including a clutch. A controller may be configured to disengage the clutch when the work vehicle speed is less than a speed threshold and when the service brake is actuated, whereby the work vehicle is brought to a stop. In embodiments, the driving assistance system may obtain a slope value representative of the slope of a surface on which the work vehicle has stopped and may automatically activate the parking brake when the clutch has been disengaged and the slope value is greater than a slope threshold. The driving assistance system may reengage the clutch when the service brake is released, while maintaining the parking brake activated. The parking brake may be automatically deactivated when a predetermined delay after release of the service brake has elapsed. The predetermined delay may be greater than the time required to reengage the clutch, or when reengagement of the clutch has reached a predetermined value.

(7) The work vehicle may be an agricultural vehicle or a construction vehicle, such as a tractor, a mower or the like (a groundcare vehicle), a telehandler, etc.

- (8) A work vehicle comprises a service brake, which may be actuated by a driver of the work vehicle and is configured, upon actuation, to decelerate the work vehicle, i.e. to decrease its speed with respect to the ground. Eventually, the work vehicle may come to a complete stop, in which case a continued actuation of the service brake would keep the vehicle stationary.
- (9) In contrast, the parking brake is configured to maintain the vehicle stopped without a continued actuation thereof. For instance, the parking brake may be configured to be switched between an activated position and a deactivated position, and able to be left in the activated position without further actuation thereof, as opposed to the service brake which is released as soon as actuation thereof is released. Besides, the parking brake may not be able to decelerate the vehicle.
- (10) A service brake and a parking brake are each known per se in the art and both are conventionally provided on vehicles. The parking brake is configured to be automatically activated and deactivated, and thus may be referred to as an assisted parking brake or an automatic parking brake.
- (11) The vehicle has a transmission: torque between the driving power source (e.g. engine, electric motor or the like) and the wheels is transmitted through the transmission. A clutch is to be disengaged when the driver wants to change the shift, and then reengaged.
- (12) As mentioned above, the controller is configured to disengage the clutch when the work vehicle speed is less than a speed threshold and the service brake is actuated. Disengaging the clutch when speed gets low is usual in vehicles in order to prevent stalling, but is generally performed by pressing a dedicated pedal. In contrast, in the above driving assistance system, the controller disengages the clutch irrespective of whether the driver actuates the clutch pedal. Therefore, the vehicle may be stopped by operating only the service brake actuator. Likewise, the controller is configured to reengage the clutch when the service brake is released, so that the driver may resume moving the vehicle by operating only the service brake actuator. Thus stopping and restarting of the work vehicle are facilitated.
- (13) In order to prevent rollback on slopes, the controller is configured to detect a slope situation, or more precisely to obtain a slope value representative of the slope of a surface on which the vehicle has stopped. The slope value may indicate directly the slope as an angle, or may be expressed in another unit which, after a calibration known per se, would represent the slope.
- (14) If the slope is greater than a slope threshold, i.e. if there is a risk of rollback, the parking brake is automatically activated, and is automatically deactivated after the driver has released the service brake actuator, specifically when a predetermined delay after release of the service brake has elapsed, the predetermined delay being greater than the time required to reengage the clutch, or when reengagement of the clutch has reached a predetermined value. Reengagement of the clutch measures the ability of the clutch to transmit torque between an input and an output thereof.
- (15) Therefore, although the service brake has been released, the parking brake is maintained until the clutch has been reengaged at least to a certain value, which is calibrated as desired to counter the tendency of the work vehicle to rollback.
- (16) As a consequence, the driving assistance system judiciously couples operation of the service brake, the clutch and the parking brake in order to enable the driver to safely drive the work vehicle, with limited or no uncontrolled movements, while operating the work vehicle in a simple manner (in this case, using only the service brake actuator) and while relying on existing components of the work vehicle, such as classical service brake, clutch and parking brake, thus limiting the cost and complexity increase.
- (17) In the present disclosure, although the example of rollback is given, this notion may be generalized to encompass any movement of the work vehicle in the direction of the slope and possibly contrary to the desired driving direction, or more generally to any uncontrolled movement of the work vehicle due to gravity.
- (18) Optionally, the controller is configured to determine the slope value based on a stroke of a service brake actuator or on a pressure within the service brake. In these embodiments, a dedicated

slope sensor is not needed, so that complexity and cost of the work vehicle is not increased. The stroke of the service brake actuator or a pressure within the service brake (e.g. a hydraulic pressure when the service brake includes a hydraulic mechanism) are representative of the driver's force necessary to bring the work vehicle to a stop on that slope and to maintain the work vehicle stopped. The greater the slope, the greater the stroke or pressure.

(19) Optionally, the work vehicle further comprises an angle sensor configured to output information representative of an angle of the work vehicle with respect to horizontal, and wherein the controller is configured to determine the slope value based on an output of the angle sensor. In these embodiments, the angle sensor may provide a more accurate measure of the slope, and possibly an indication of the direction of the slope (e.g. forwards or backwards). The angle sensor may be used alone or as a complement to the above-mentioned stroke or pressure.

(20) Optionally, the controller is configured to determine that the service brake has been released based on a stroke of a service brake actuator or on a pressure within the service brake. For instance, the controller may determine that the service brake has been released when the stroke or pressure is less than a threshold.

(21) Optionally, the predetermined value is a value of a pressure within an engagement mechanism of the clutch. The predetermined value is the threshold at which the parking brake is deactivated. For instance, if the engagement mechanism of the clutch includes hydraulic components, the pressure may be the hydraulic pressure.

(22) Optionally, the driving assistance system further comprises an input device configured to enable a user to input settings for the controller. The input device may comprise a button, a lever, a switch, a touch panel, a communication device configured to connect to an external terminal, etc.

(23) Optionally, the input device is configured to enable a user to turn on/off said automatic activation and deactivation of the parking brake. Therefore, the user (or driver) may turn on the automatic activation/deactivation of the parking brake when the vehicle is driven on a sloped ground, and may turn off said automatic activation/deactivation when the driver does not need it, for instance when the vehicle is driven on roads and the driver may suddenly brake without being on a slope, or when the vehicle is driven with a loader. In the latter case, the driver may look for reactivity despite frequent brake actuations; thus, in these circumstances, automatically activating the parking brake may not be suitable. Therefore, the driving assistance system is able to adapt to all driving situations.

(24) Optionally, the input device is configured to enable a user to calibrate the slope threshold. The slope threshold is the value from which the parking brake is activated when the vehicle is stopped. Therefore, the user may adapt the automatic behavior of the work vehicle to his needs and driving habits.

(25) Optionally, the input device is configured to set the slope threshold at a value chosen by the user within a list of preset values. Therefore, calibration is easy and corresponds to safe and standard settings.

(26) Optionally, alternatively or in addition, the input device is configured to record, as the slope threshold, a value corresponding to an actuation of a service brake actuator. In particular, when the slope value is determined based on the stroke of a service brake actuator or on a pressure within the service brake, the driver may calibrate the slope threshold by calibrating directly the detected quantity, e.g. the stroke or the pressure respectively. By actuating the service brake actuator to the desired threshold, the service brake actuator is set to a given stroke, and a pressure within the service brake is set to a given value, and the stroke and/or the pressure value may be recorded to identify the desired slope threshold. Therefore, the driving assistance system is highly configurable and the settings may optionally be input in a plurality of manners.

(27) All in all, the driving assistance system not only enables the driver to safely drive the work vehicle in a simple and inexpensive manner, but also provides the work vehicle with much versatility and flexibility, such that every driver may perform the desired working tasks, depending

on the work implement used, as per his best convenience.

(28) The present disclosure is further related to a driving assistance method for a work vehicle comprising a service brake configured to decelerate the work vehicle upon actuation of the service brake, a parking brake and a transmission including a clutch, the driving assistance method comprising: disengaging the clutch when the work vehicle speed is less than a speed threshold and the service brake is actuated, whereby the work vehicle is brought to a stop; obtaining a slope value representative of the slope of a surface on which the work vehicle has stopped; automatically activating the parking brake when the clutch has been disengaged and the slope value is greater than a slope threshold; reengaging the clutch when the service brake is released, while maintaining the parking brake activated; and automatically deactivating the parking brake when a predetermined delay after release of the service brake has elapsed, the predetermined delay being greater than the time required to reengage the clutch, or when reengagement of the clutch has reached a predetermined value.

(29) The driving assistance method may be carried out by the above-described driving assistance system, and may further include further steps corresponding to some or all of the above-described optional features.

(30) The present disclosure is further related to a computer program including instructions for executing the steps of the above described driving assistance method when said program is executed by a computer. This program may use any programming language and take the form of source code, object code or a code intermediate between source code and object code, such as a partially compiled form, or any other desirable form.

(31) The present disclosure is further related to a recording medium readable by a computer and having recorded thereon a computer program including instructions for executing the steps of the above described driving assistance method. The recording medium may be any entity or device capable of storing the program. For example, the medium may include storage means such as a ROM, for example a CD ROM or a microelectronic circuit ROM, or magnetic storage means, for example a diskette (floppy disk) or a hard disk.

(32) Alternatively, the recording medium may be an integrated circuit in which the program is incorporated, the circuit being adapted to execute the method in question or to be used in its execution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention and advantages thereof will be better understood upon reading the detailed description which follows, of embodiments given as non-limiting examples. This description refers to the appended drawings, wherein:

(2) FIG. 1 is a diagram illustrating a work vehicle according to an embodiment; and

(3) FIG. 2 is a flowchart illustrating a driving assistance method according to an embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(4) A work vehicle **10** is diagrammatically illustrated in FIG. 1. Systems of the work vehicle **10** which are not relevant to the present disclosure are omitted and may be provided as known in the art. For instance, as a work vehicle, the work vehicle **10** may comprise a work implement, a trailer, etc.

(5) The work vehicle **10** comprises wheels **12**, generally two front wheels and two rear wheels. More or less wheels may be provided.

(6) The work vehicle **10** comprises a service brake **20**. The service brake **20** is configured to decelerate the work vehicle **10** upon actuation. To this end, the service brake **20** may comprise a service brake actuator **22** and a service brake mechanism **24**. The service brake actuator **22** is

actuable by a driver of the work vehicle **10**. The service brake mechanism **24** receives an input from the service brake actuator **22** and acts on the wheels **12** (or an axle, or a shaft connected to the wheels **12**) accordingly to decelerate the work vehicle **10**. The service brake **20** itself may be implemented as known per se in the art. For instance, the service brake actuator **22** may comprise a pedal. For instance, the service brake mechanism **24** may comprise a hydraulic system.

(7) The work vehicle **10** further comprises a parking brake **26** distinct from the service brake **20**. The parking brake **26** is configured, when activated, to keep the work vehicle **10** stopped, irrespective of the actuation of any other system of the work vehicle **10**. The parking brake **26** is actuable by a user of the work vehicle **10**. The parking brake **26** itself may be implemented as known per se in the art.

(8) The work vehicle **10** further comprises a driving power source **36** such as an engine, an electric motor or the like. The driving power source **36** may include one or several engines, electric motors or the like, or a combination thereof. The work vehicle **10** is provided with a transmission **28**, such as a power shift transmission, configured to transmit torque from the driving power source **36** to the wheels **12**. The power shift transmission, as opposed to a continuous variable transmission, enables the user to select a shift among a plurality of discrete shifts, which may be implemented by corresponding gears, as known per se in the art.

(9) In particular, the transmission **28** includes a clutch **30**. The clutch **30** may comprise an engagement mechanism **32**, configured to engage or disengage an input component connected to the driving power source **36** to/from an output component connected to the wheels **12**. When the clutch **30** is engaged (or closed), torque is transmitted from the driving power source **36** to the wheels **12** via the transmission **28**. Conversely, when the clutch **30** is disengaged (or open), torque from the driving power source **36** is not transmitted to the wheels **12** via the transmission **28**.

(10) The clutch **30** may comprise a clutch actuator **34**, actuable by a user of the work vehicle **10**, and configured to transmit an engagement or disengagement signal to the engagement mechanism **32**. For instance, the clutch actuator **34** may be a clutch pedal.

(11) The work vehicle **10** is provided with a driving assistance system which comprises a controller **40**. The controller **40** may be a dedicated controller or may be common to other systems of the work vehicle **10**.

(12) As illustrated in FIG. 1, the controller **40** may be connected to the service brake actuator **22**. For instance, the controller **40** may be configured to receive a value indicative of the stroke of the service brake actuator **22**. Based on that stroke, the controller **40** may be configured to determine a slope on which the work vehicle **10** is stopped: for instance, the driver needs to apply a relatively great force to the service brake actuator **22**, hence to move it along a relatively great stroke, in order to maintain the work vehicle **10** stopped on a steep slope. Conversely, the driver needs to apply only a relatively small force to the service actuator **22**, hence to move it along a relatively short stroke, in order to maintain the work vehicle **10** stopped on a gentle slope.

(13) As illustrated in FIG. 1, the controller **40** may be connected to the service brake mechanism **24**. For instance, the controller **40** may be configured to receive a value indicative of the pressure within the service brake **20**, in particular, within the service brake mechanism **24** or at an interface between the service brake mechanism **24** and the service brake actuator **22**. Based on that pressure, the controller **40** may be configured to determine a slope on which the work vehicle **10** is stopped: for instance, the driver needs a relatively high braking force of the service brake mechanism **24**, hence a relatively high pressure within the service brake **20**, in order to maintain the work vehicle **10** stopped on a steep slope. Conversely, the driver needs a relatively low braking force of the service brake mechanism **24**, hence a relatively low pressure within the service brake **20**, in order to maintain the work vehicle **10** stopped on a gentle slope.

(14) The connection between the controller **40** and at least one of the service brake actuator **22** and the service brake mechanism **24** allows inter alia the controller to determine a slope value, i.e. a value representative of the slope of a surface on which the vehicle has stopped. Based on the same

principles, the controller **40** may be configured to determine that the service brake **20** has been released based on a stroke of the service brake actuator **22** and/or on a pressure within the service brake **20**.

(15) The controller **40** may be connected to the parking brake **26**. The controller **40** may be configured to automatically activate or deactivate the parking brake **26** under certain circumstances, as will be detailed hereinafter. Besides, for instance, the controller **40** may be configured to automatically activate the parking brake **26** when the driver leaves his seat.

(16) The controller **40** may be connected to the clutch **30**, in particular to the engagement mechanism **32**. The controller **40** is configured to automatically engage or disengage the clutch **30** under certain circumstances, as will be detailed hereinafter. Note that, in this embodiment, the connection between the controller **40** and the engagement mechanism **32** bypasses the clutch actuator **34**: thus, when engagement or disengagement of the clutch **30** is managed by the controller **40**, no operation of the clutch actuator **34** may be required, which facilitates operation of the work vehicle **10** for the driver.

(17) The work vehicle **10** may comprise a speed sensor **42**, configured to determine a speed of the work vehicle **10**. Direct and indirect speed sensors are known in the art. The speed sensor **42** may be configured to output the speed of the work vehicle **10** or to output a comparison result between the speed of the work vehicle **10** and one or more given values, e.g. thresholds. The controller **40** may be connected to the speed sensor **42** in order to determine whether a speed of the work vehicle **10** is less than a speed threshold. For instance, the speed threshold may be less than 5 kilometers per hour (kph), preferably less than 2 kph, preferably less than 1 kph.

(18) The work vehicle **10** may comprise an angle sensor **44** configured to output information representative of an angle of the work vehicle with respect to horizontal. Angle sensors for vehicles are known per se in the art. As indicated above, the slope value may be determined by the controller **40** through an output of any one of the service brake actuator **22**, the service brake mechanism **24** and the slope sensor **44**, or a combination thereof in order to make the determination more accurate and/or reliable.

(19) The work vehicle **10** may comprise an input device **46** for a user to input settings for the controller **40**. Although the input device **46** is diagrammatically illustrated as a touch panel, the input device **46** may be any other means enabling the user to tune settings of the controller **40**, as mentioned above. Preferably, the input device **46** is such that it is accessible to a standard user or driver of the work vehicle **10**, and not only to shops or dealers.

(20) In this embodiment, the controller **40** has the hardware configuration of a computer: it comprises inter alia a processor, a memory and communications means with the other devices to which it is connected, as described above and enabling the controller **40** to acquire values to these devices. The memory forms a recording medium which may be read by the processor and on which a computer program is stored, the computer program comprising instructions for carrying out the driving assistance method **100** which will be described with reference to FIG. 2.

(21) FIG. 2 is a flowchart illustrating the driving assistance method **100** (hereinafter “the method”) according to an embodiment. As shown in frame E1 of FIG. 2, as an initial situation, it is assumed that the work vehicle **10** is driving on an upward slope and that the driver wants to stop on that slope. The driving assistance method **100** is to be applied in real time, repeatedly during operation of the work vehicle **10**.

(22) At step **102**, the method evaluates whether the service brake **20** has been actuated. If the service brake **20** has not been actuated (NO), the method terminates. Conversely, if the service brake **20** has been actuated (YES), it should result in a speed decrease of the work vehicle **10** as shown in frame E2, and the method proceeds to step **104**.

(23) At step **104**, the method determines whether the work vehicle speed is less than a speed threshold. This determination may be carried out by the controller **40**, e.g. based on the output of the speed sensor **42**. If the work vehicle speed is greater than the threshold (NO), the method

repeats step **104**, as the work vehicle **10** is still moving too quickly to stop. In other words, the method waits for the work vehicle **10** to be close to stopping, or to be stopped altogether. When the work vehicle **10** has stopped or is nearly stopped, for example, when the work vehicle speed is less than the speed threshold (YES), the method proceeds to step **106**.

(24) At step **106**, since the work vehicle speed is less than a speed threshold and the service brake is actuated, the controller **40** disengages the clutch **30**. Under these circumstances, the work vehicle **10** is brought to a stop. This situation is illustrated in frame E3. Then, the method proceeds to step **108**.

(25) At step **108**, the controller **40** obtains a slope value representative of the slope of a surface on which the work vehicle **10** has stopped. The slope value may be obtained by the controller as discussed above. Since the ways for obtaining the slope value are independent of the clutch **30**, the slope value may be obtained before disengaging the clutch at step **106**, simultaneously thereto, or afterwards as illustrated.

(26) After obtaining the slope value, the controller **40** determines whether the slope value is greater than a slope threshold at step **110**. If the slope value is greater than the slope threshold at step **110** (YES), the method proceeds to step **112**. Otherwise (NO), the method proceeds directly to step **114**.

(27) When the clutch has been disengaged at step **106** and the slope value obtained at step **108** is greater than a slope threshold, as determined in step **110**, the controller **40** automatically activates the parking brake **26** at step **112**, as illustrated in frame E4. In other words, when the slope is steeper than the slope threshold, it is determined that the parking brake **26** will be helpful for the work vehicle **10** to resume moving, especially moving forward, and the parking brake **26** is automatically activated by the controller **40**. The condition on the clutch **30** being disengaged ensures that the parking brake is not activated while the clutch **30** transmits torque from the driving power source **26** to the wheels **12**, which is likely to lead to stalling.

(28) Conversely, when the slope value is less than the slope threshold, i.e. the slope is relatively gentle, the controller **40** may not automatically activate the parking brake **26**, because the parking brake **26** may not be required for the work vehicle **10** to resume moving, and may even slow down the operation of the work vehicle **10**.

(29) After activating the parking brake **26**, or after determining that the slope value is less than the slope threshold, the method proceeds to step **114**. At step **114**, the method determines whether the service brake **20** has been released. That is, the method determines whether the driver wants the work vehicle **10** to resume moving. Service brake release detection may take place as described above. An arbitrary long time may elapse between steps **112** and **114**, insofar as the parking brake **26** has been activated or the user keeps the service brake **20** actuated. As long as the service brake has not been released (NO at step **114**), step **114** is repeated.

(30) Otherwise, when the service brake has been released (YES at step **114**), the clutch **30** is reengaged by the controller **40** at step **116**, while maintaining the parking brake **26** activated. This is illustrated in frame E5.

(31) Then, at step **118**, the method checks whether the parking brake was activated. If the parking brake was not activated (NO), then the method terminates, as the work vehicle **10** is capable of moving as soon as the clutch **30** has been reengaged.

(32) Otherwise, if the parking brake **26** was activated (YES), for instance at step **112**, the controller **40** does not deactivate the parking brake immediately, but the method proceeds to step **120**, in which the controller **40** waits for a predetermined delay after release of the service brake **20**, the predetermined delay being greater than the time required to reengage the clutch **30**, or waits for reengagement of the clutch **30** to reach a predetermined value.

(33) For instance, said predetermined value represents a degree of reengagement of the clutch **30**, as reengagement is a continuous process between two extreme states, namely the clutch **30** being fully engaged and the clutch **30** being fully disengaged. For example, if the engagement mechanism **32** of the clutch **30** includes hydraulic components or the like, the predetermined value may be a

value of a pressure within the engagement mechanism **32**.

(34) The predetermined delay may be tuned to be long enough to enable reengagement of the clutch **30**, but short enough in order to prevent stalling of the work vehicle **10**. The starting point of the predetermined delay is the release of the service brake **20**, e.g. when the service brake actuator **22** is released up to a certain level (i.e. the stroke of the service brake actuator **22** reaches a threshold), or when a pressure within the service brake **20** reaches a threshold. In a variant, the starting point of the predetermined delay may be when the reengagement of the clutch **30** reaches a predetermined value.

(35) Either the predetermined delay or the predetermined value may be constant or variable as a function of other parameters, such as the slope value, the parking brake oil temperature where applicable, etc.

(36) When the predetermined delay after release of the service brake **20** has elapsed, or when reengagement of the clutch **30** has reached the predetermined value, the driving assistance method proceeds to step **122**, in which the controller **40** deactivates the parking brake **26**. This situation is illustrated by frame E6. Afterwards, the driving assistance method **100** terminates, as the work vehicle **10** is able to resume moving with a limited or no rollback, as shown in frame E7.

(37) Although the driving assistance method **100** has been described with reference to the controller **40** and the embodiment of the work vehicle **10** as detailed in FIG. **1**, the driving assistance method is not limited to that embodiment and may be carried out by other means.

(38) In some embodiments, the input device **46** enables the user to turn on/off said automatic activation and deactivation of the parking brake **26**. To this end, the input device **46** may include a physical switch or the like, or a virtual switch which may be operated by the touch panel or another control present in the work vehicle (joystick, etc.).

(39) FIG. **2** represents a case in which the automatic activation and deactivation of the parking brake **26** is turned on. When it is turned off, the controller **40** may ignore steps **110** and **112** so that the parking brake **26** is not automatically activated, irrespective of the slope value. Step **118** may be ignored or may be carried out, but since the parking brake was not activated (NO in step **118**), the deactivation thereof is not necessary, and steps **120** and **122** are not carried out in either case.

(40) Step **110** relies on a comparison with a slope threshold. In some embodiments, the slope threshold may be constant, e.g. set by the manufacturer. In other embodiments, the user may be able to calibrate the slope threshold, e.g. through the input device.

(41) For instance, the input device **46** may give the user options within a list of preset values (e.g. light, medium and hard sensitivity). The values may themselves be constant, e.g. set by the manufacturer. Thus, the user may select the slope threshold which is the best adapted to his way of driving the work vehicle **10** and to his operational needs.

(42) In another example, the slope threshold may be selected not among discrete values but within a continuous range, for a more accurate calibration. The desired slope threshold may be input manually into the input device **46**. However, the input device **46** may otherwise be configured to record, as the slope threshold, a value corresponding to an actuation of the service brake actuator **22**. As detailed above, a value corresponding to an actuation of the service brake actuator **22** may be a stroke value or a pressure value and is representative of a slope.

(43) Thus, for example, the user may perform a calibration operation of the slope threshold by starting a calibration phase on the input device **46**, actuating the service brake actuator **22** to a desired level (e.g. stroke or pressure), and to make the input device **46** record the value at which the service brake actuator **22** is being actuated. Optionally, the input device **46**, e.g. in case of a touch panel, or another device, may display the recorded value, or even the current value to which the service brake actuator **22** is being actuated, to facilitate calibration.

(44) Although the present disclosure refers to specific exemplary embodiments, modifications may be provided to these examples without departing from the general scope of the invention as defined by the claims. For instance, although FIG. **1** illustrates some connections between the different

components, the actual connections may be carried out in a different way in order to yield the same functions. As another example, although FIG. 2 presents steps of the driving assistance method in a certain order, the skilled person may modify this order in a technically appropriate manner while still carrying out the relevant steps of the method. More generally, individual characteristics of the different illustrated/mentioned embodiments may be combined in additional embodiments. Therefore, the description and the drawings should be considered in an illustrative rather than in a restrictive sense.

Claims

1. A driving assistance system for a work vehicle comprising a service brake configured to decelerate the work vehicle upon actuation of the service brake, a parking brake and a transmission including a clutch, the driving assistance system comprising a controller configured to: disengage the clutch when the work vehicle speed is less than a speed threshold and the service brake is actuated, whereby the work vehicle is brought to a stop; obtain a slope value representative of a slope of a surface on which the work vehicle has stopped; automatically activate the parking brake when the clutch has been disengaged and the slope value is greater than a slope threshold; reengage the clutch when the service brake is released, while maintaining the parking brake activated; and automatically deactivate the parking brake when a predetermined delay after release of the service brake has elapsed, the predetermined delay being greater than the time required to reengage the clutch, or when reengagement of the clutch has reached a predetermined value.
2. The driving assistance system of claim 1, wherein the controller is configured to determine the slope value based on a stroke of a service brake actuator or on a pressure within the service brake.
3. The driving assistance system of claim 1, wherein the work vehicle further comprises an angle sensor configured to output information representative of an angle of the work vehicle with respect to horizontal, wherein the controller is configured to determine the slope value based on an output of the angle sensor.
4. The driving assistance system of claim 1, wherein the controller is configured to determine that the service brake has been released based on a stroke of a service brake actuator or on a pressure within the service brake.
5. The driving assistance system of claim 1, wherein the predetermined value is a value of a pressure within an engagement mechanism of the clutch.
6. The driving assistance system of claim 1, further comprising an input device configured to enable a user to input settings for the controller.
7. The driving assistance system of claim 6, wherein the input device is configured to enable a user to turn on/off said automatic activation and deactivation of the parking brake.
8. The driving assistance system of claim 6, wherein the input device is configured to enable a user to calibrate the slope threshold.
9. The driving assistance system of claim 8, wherein the input device is configured to set the slope threshold at a value within a list of preset values.
10. The driving assistance system of claim 8, wherein the input device is configured to record, as the slope threshold, a value corresponding to an actuation of a service brake actuator.
11. A driving assistance method for a work vehicle comprising a service brake configured to decelerate the work vehicle upon actuation of the service brake, a parking brake and a transmission including a clutch, the driving assistance method comprising: disengaging the clutch when the work vehicle speed is less than a speed threshold and the service brake is actuated, whereby the work vehicle is brought to a stop; obtaining a slope value representative of a slope of a surface on which the work vehicle has stopped; automatically activating the parking brake when the clutch has been disengaged and the slope value is greater than a slope threshold; reengaging the clutch when the service brake is released, while maintaining the parking brake activated; and automatically

deactivating the parking brake when a predetermined delay after release of the service brake has elapsed, the predetermined delay being greater than the time required to reengage the clutch, or when reengagement of the clutch has reached a predetermined value.

12. The driving assistance method of claim 11, further comprising a computer program including instructions for executing the steps of the method of claim 11 when said program is executed by a computer.

13. The driving assistance method of claim 11, further comprising a recording medium readable by a computer and having recorded thereon a computer program including instructions for executing the steps of the method of claim 11.

14. The driving assistance method of claim 11, further comprising determining the slope value based on a stroke of a service brake actuator or on a pressure within the service brake.

15. The driving assistance method of claim 11, further comprising outputting information representative of an angle of the work vehicle with respect to horizontal, wherein the slope value is determined based on an output of the angle sensor.

16. The driving assistance method of claim 11, further comprising determining that the service brake has been released based on a stroke of a service brake actuator or on a pressure within the service brake.

17. The driving assistance method of claim 11, wherein the predetermined value is a value of a pressure within an engagement mechanism of the clutch.

18. The driving assistance method of claim 11, further comprising enabling a user to input settings for a controller controlling the driving assistance method.

19. The driving assistance method of claim 11, further comprising enabling a user to turn on/off automatic activation and deactivation of the parking brake.

20. The driving assistance system of claim 11, further comprising enabling a user to calibrate the slope threshold.
